

SHEILLA MUMBI MACHARIA

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PROFESSIONAL SUMMARY

Data Science practitioner with a background in Applied Mathematics and hands-on experience in predictive modeling, analytics, and visualization. Skilled in Python, Tableau, and Streamlit for deriving actionable insights and building data-driven applications. Passionate about turning raw data into meaningful stories that drive smarter decisions, with proven success in projects spanning health prediction, retail analytics, and aviation risk assessment.

TECHNICAL SKILLS

Languages: Python, SQL, HTML/CSS

Libraries/Frameworks: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit

Tools: Excel, Tableau, Power BI, Jupyter Notebook, Git

Core Skills: Data Cleaning, Feature Engineering, Predictive Modelling, Data Visualization, Business Intelligence

PROJECTS

SyriaTel Retention & Revenue Insights | *Python, Scikit-learn* | [Repository](#)

- Consolidated and validated telecom customer datasets to ensure consistency across tenure, usage, and service records.
- Conducted exploratory analysis to identify churn patterns and quantify potential revenue-at-risk segments.
- Highlighted key commercial drivers of churn, providing actionable insights for targeted retention strategies.

Supermarket Sales Analysis and Visualization Case Study | *Python (Pandas, Matplotlib, Seaborn)*; [Notebook](#)

- Conducted an end-to-end analysis of store performance data to assess the impact of store size, customer traffic, and product range on sales.
- The analysis revealed that medium-sized stores achieved the highest sales efficiency (sales per square foot) despite lower traffic.
- Delivered actionable insights and recommendations for optimizing layout, promotions, and inventory strategy across stores.

Movie Financial Performance Analysis | *Python, Tableau* | [Repository](#)

- Consolidated revenue, cost, and ROI data across multiple projects to assess financial performance.
- Validated metrics to detect discrepancies and highlighted high-return productions and budget inefficiencies.
- Presented insights in executive-ready dashboards to inform strategic investment decisions.

Aviation Risk Insights Project | *Tableau, Python, Pandas* | [Repository](#) and [Dashboard](#)

- Analyzed over a century of global aviation accident data to identify trends, severity patterns, and operational risk factors.
- Analysis revealed that a small subset of aircraft models accounted for most severe accidents, highlighting critical safety focus areas.
- Designed an interactive Tableau dashboard to help aviation authorities and airlines visualize risk patterns and enhance safety measures.

SQL Labs & Commercial KPI Practice | [GitHub](#)

- Applied SQL to join, filter, and aggregate multi-table datasets.
- Calculated KPIs such as customer churn rate, segment revenue, and renewal percentages.
- Practiced reconciling aggregated outputs against source-level records to ensure data accuracy.

EDUCATION

Bachelor of Science in Applied Mathematics

University of Nairobi | Expected Graduation: 2026

Moringa School Data Science Bootcamp (Ongoing – February 2026)

Intensive hands-on program in Python, Machine Learning, and Data Visualization, culminating in real-world capstone projects (SyriaTel Customer Churn Prediction, Aviation Risk Insights).

CERTIFICATIONS

Data Science Program – eMobilis Technological Institute

Comprehensive training in data science fundamentals, covering Python programming, data wrangling, exploratory data analysis, and introductory machine learning. Completed hands-on projects involving real-world datasets to develop skills in data cleaning, feature engineering, and insight generation for decision-making.

Python for Data Analysis – FreeCodeCamp

Focused training in Python for data analysis, including NumPy, pandas, and data visualization libraries. Applied techniques for data cleaning, transformation, and exploratory analysis to extract meaningful insights from structured datasets.

Machine Learning & Data Visualization – Kaggle

Hands-on learning in machine learning workflows and data visualization best practices using real-world datasets. Built predictive models, evaluated performance metrics, and created visual dashboards to communicate insights effectively.

LANGUAGES

English (Fluent) | Swahili (Fluent)